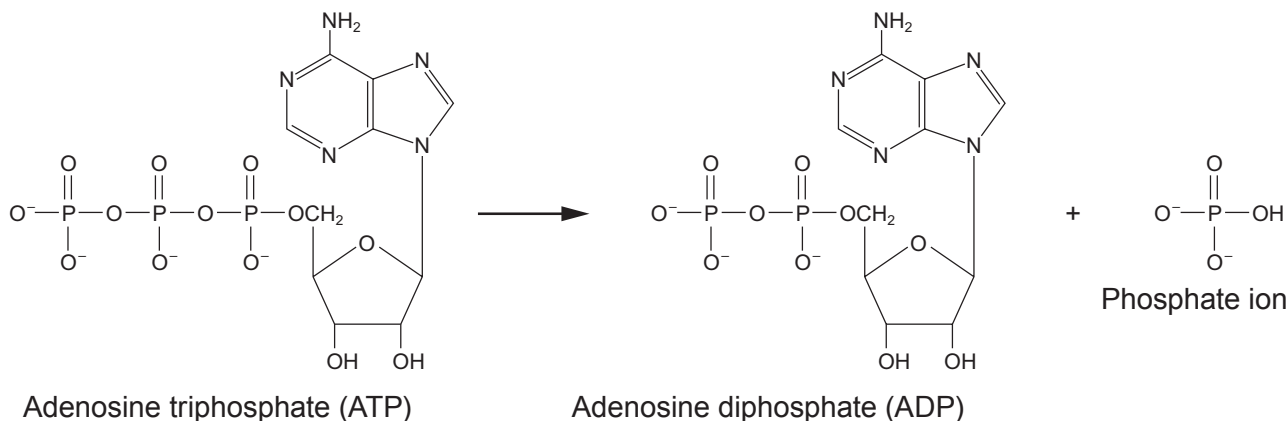


3. Adenosine triphosphate (ATP) belongs to a group of molecules that are made from nucleotides. ATP is an unstable molecule which possesses two high energy phosphate bonds and is often referred to as the universal energy currency. The diagram below shows the conversion of ATP into ADP.



- (a) (i) State the parts of the molecule that are common to DNA, RNA and ATP nucleotides. [1]

.....

.....

- (ii) State what is meant by the term *universal energy currency*. [1]

.....

.....

.....

- (iii) State whether the reaction shown above is exergonic or endergonic. Explain your answer. [2]

.....

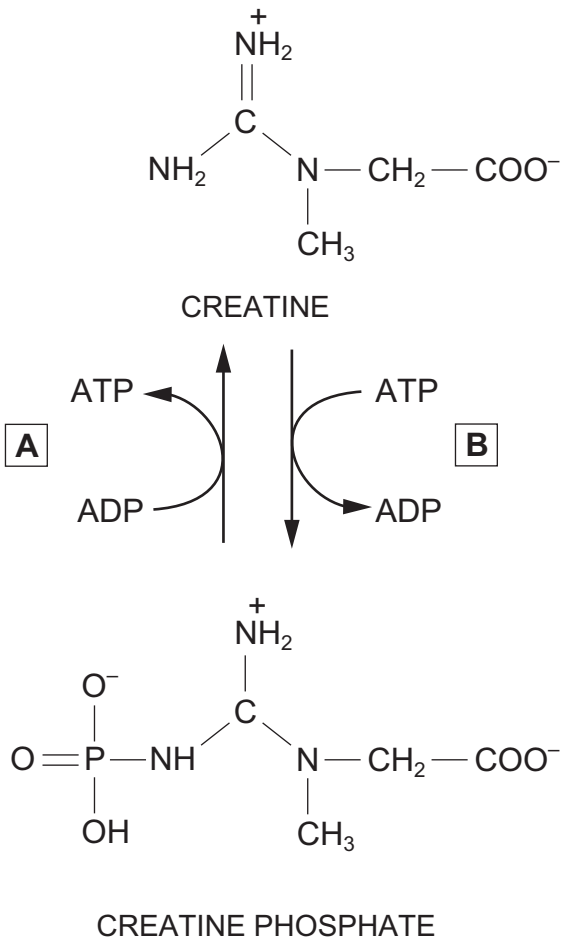
.....

- (b) It is estimated that the mass of ATP that is found within a resting human at any one time is approximately 5g, yet a human uses an estimated 50 kg of ATP each day. Calculate the percentage of the total ATP used per day that is available at any one time. [2]

Percentage of the total ATP available = %



Creatine phosphate is a naturally occurring chemical that works in partnership with ATP to increase the ability of muscle tissue to perform. It is not an energy source itself, but is used to increase the availability of ATP. It is stored inside muscle tissue.



(c) Under what circumstances would each of the reactions **A** and **B** take place? Explain your answer. [4]

